Spatial Prediction of Water Quality in Arizona Using Gaussian Process Regression

Engineering Goals

- **By testing water at just 6 Phoenix sites, this project uses AI to predict metal contamination levels across the entire Phoenix metropolitan area**
- **Measure :** Collect dissolved metal concentrations (Fe, Cr, Mn, Mo, In) from six sites across Phoenix using calibrated FAAS analysis.
- **Model :** Train a Gaussian Process regression model on those six measurements to predict concentrations at any unsampled location across the metropolitan area.
- **Quantify Uncertainty :** Produce a confidence estimate alongside every prediction, identifying where the model is reliable and where it is not.
- **Validate :** Verify model accuracy using Leave-One-Out Cross-Validation, the most rigorous approach possible dataset.

Project Origin

- Arizona water bodies have variable dissolved metal levels; some sites **exceed safe thresholds**.
- Lab analysis of water samples costs **$50–$200** per site and **takes days**.
- Most lakes, rivers, and canals remain **unmonitored** due to limited resources.
- **Current prediction methods** require dense sampling or ignore uncertainty, leaving high-risk areas unidentified.

Methods Project Design

- **Spatial Modeling Framework:** Uses Gaussian Process Regression (GPR) to create predictions of water quality at unsampled locations.
- **Data Collection**: Collected water samples from specific sites, mostly concentrated in the metropolitan Phoenix area, which is why the dataset is small but precise.
- **Validation:** Compared predictions to held-out sample points to evaluate accuracy and reliability.
- **Measured elements**: Concentrations of iron (Fe), chromium (Cr), manganese (Mn), molybdenum (Mo), and indium (In) in water samples.

Methods Why: Gaussian Process

- **Simple Explanation:** GPR predicts water quality at places we haven't measured by looking at patterns in the data we do have, and it also tells us how confident it is in each prediction.
- **Technical Explanation:** GPR treats measurements as samples from a multivariate Gaussian distribution over functions. It uses a covariance function (kernel) to model how points in space relate to each other. This allows it to interpolate data in a smooth, flexible, and non-linear way, and it naturally provides uncertainty estimates at unmeasured locations.
- **Why it's better for this project**: Gaussian processes work well with sparse, uneven data, capture complex spatial patterns, and provide confidence in predictions, making them more reliable than simpler regression or interpolation methods.

Methods *Mathematical Model: GP*

$$k(x, x') = \sigma_f^2 \exp\left(-\frac{||x - x'||^2}{2\ell^2}\right)$$

- **Squared Exponential (RBF) Kernel:** Measures how similar two locations are based on distance. Closer sites get a score near 1, distant sites near 0.

$$\mu_* = k_*^T (K + \sigma_n^2 I)^{-1} y$$

- **GP Posterior Mean:** Predicts the metal concentration at any unsampled location by blending the 6 known measurements weighted by their spatial similarity to that point.

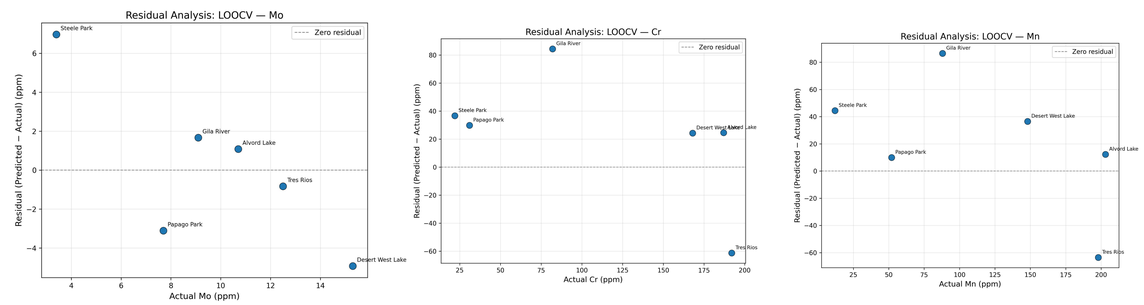
$$\sigma_*^2 = k(x_*, x_*) - k_*^T (K + \sigma_n^2 I)^{-1} k_*$$

- **GP Posterior Variance:** Quantifies prediction confidence. Low near training sites, high in unexplored areas.

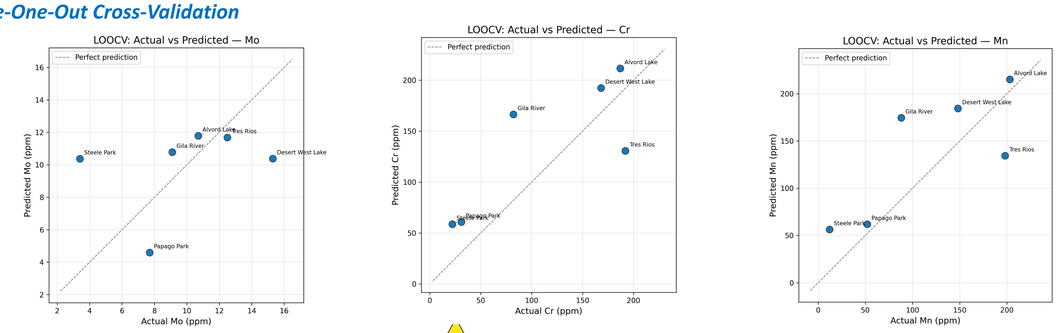
(Rasmussen and Williams)

Results

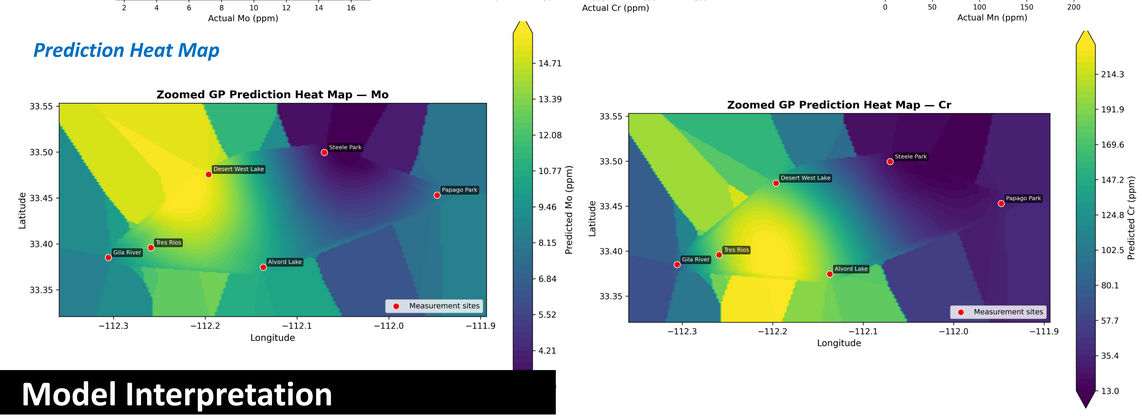
Bias or Error Patterns



Leave-One-Out Cross-Validation



Prediction Heat Map



Model Interpretation

Evidence Against Overfitting	Model Behavior and Safeguards
LOOCV errors are moderate and consistent across folds	RBF kernel enforces smooth, non-excessive variation
Residuals are centered around zero with no strong bias	Hyperparameter optimization penalizes overly complex fits
Predictions remain stable when individual points are removed	Uncertainty increases in data-sparse regions, preventing overconfidence
No extreme error spikes in any single validation fold	Predictions revert toward the mean far from sampled sites

Findings and Significance

- **Findings from water quality spatial prediction were analyzed and interpreted as follows:**
- **Due to the small sample size, results should be interpreted as a proof of concept rather than a definitive spatial analysis**

1. **Prediction accuracy:**
   - ✔ LOOCV residual maps show that most predicted values closely match actual measurements, confirming the Gaussian process captures spatial trends effectively.
   - ✔ Highest spatial variation occurs near metropolitan Phoenix; predictions smooth out beyond ~30 km, showing limits of extrapolation.
   - ✔ Gaussian process provides uncertainty estimates, helping identify which predictions are more or less trustworthy.
   - ✔ Residual patterns do not show major systematic bias, suggesting the model is not consistently over- or under-predicting.
2. **Limitations:**
   - ✔ Our dataset is very small and concentrated in metropolitan Phoenix, so predictions beyond this area are extrapolated and less reliable.
   - ✔ Individual anomalous sites (e.g., Steele Park) can disproportionately influence local predictions and hyperparameter selection.

Methods *Data Collection and Analysis*

- **Sampling Sites:** Six sites across metropolitan Phoenix, selected to span diverse land uses and potential contamination sources.

Site	Location	Context
Alvord Lake	West Phoenix	Urban lake
Desert West Lake	NW Phoenix	Urban lake
Steele Indian School Park	Central Phoenix	Urban park
Gila River (Estrella)	SW Phoenix	River system
Papago Park	East Phoenix	Near desert
Tres Rios Wetlands	West Phoenix	Constructed wetland

- **Analytical Method:** Flame Atomic Absorption Spectroscopy (FAAS).
- **FAAS Step 1:** A liquid sample is sprayed into an air-acetylene flame (~2,300°C).
- **FAAS Step 2:** Atoms in the flame absorb light from an element-specific hollow cathode lamp.
- **FAAS Step 3**: The amount of light absorbed is directly proportional to the metal concentration.
- **Beer-Lambert Law:** converts measured absorbance into an exact metal concentration.

$$A = \varepsilon \cdot l \cdot c$$

Measured Concentrations

Metal	Range (mg/L)	Key Finding
Iron (Fe)	26.9 – 73.2	Highest concentrations; required dilution
Chromium (Cr)	0.019 – 0.198	Strong west → east gradient
Manganese (Mn)	0.003 – 0.211	Co-varies with Cr
Molybdenum (Mo)	0.003 – 0.016	Co-varies with Cr and Mn
Indium (In)	<LOD – 0.019	Below detection at half the sites

- **Key Pattern Discovered:** Cr, Mn, and Mo are highest in western Phoenix and decrease moving east, suggesting a shared industrial or geological source in the western metro area.

Next Steps and Conclusion

- **Increase sample size:** Expand data collection beyond metropolitan Phoenix to improve spatial coverage and reduce extrapolation.
- **Add temporal sampling:** Collect seasonal or multi-year measurements to model time-based variation in water quality.
- **Validation:** Compared predictions to held-out sample points to evaluate accuracy and reliability.
- **Incorporate environmental covariates**: Integrate land use, soil composition, elevation, groundwater flow, or infrastructure data to improve predictive accuracy.
- **Multi-element modeling:** Implement multi-output Gaussian processes to capture correlations between iron, chromium, manganese, molybdenum, and indium.

Key References

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